Package 'LFQPCA'

November 4, 2026

Type Package

Title Longitudinal Functional Quantile Principal Component Analysis

Version 0.1.0

Description Longitudinal functional quantile principal component analysis (LFQPCA) decomposes the conditional quantiles of curves recorded at repeated visits into a population surface, persistent subject trajectories with functional intercepts and slopes in time, and transient visit deviations that carry no trend within a subject. Longitudinal functional quantile principal component regression (LFQPCR) relates the quantiles of a scalar response to the two fitted components through separate coefficient functions. Estimation minimizes a subject-balanced check loss over spline bases. The package includes prediction for new subjects, cross-validation, a subject bootstrap and the data-generating designs of the simulation study in the accompanying paper.

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Encoding UTF-8

Depends R (>= 4.0.0)

Imports graphics, pbs, quantreg, splines, stats, utils

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`LFQPCA-package` *Longitudinal functional quantile principal component analysis and regression*

Description

Quantile decomposition of curves recorded at repeated visits into a population surface, persistent subject trajectories and transient visit deviations, and quantile regression of a scalar response on the two subject-level components. The package also generates the data of the simulation designs of the accompanying paper.

Details

Subject i is seen at visits $j = 1, \ldots, J_i$ at times T_{ij}. Each visit gives a curve $Y_{ij}(s)$ on a grid of the functional domain and, possibly, a scalar response V_{ij}. With standardized time $u_{ij} = (T_{ij} - \bar{T})/s_T$ and $H(u) = (1, u, \ldots, u^p)^\top$, the conditional τ-quantile of the curves is modeled as

$$Q_\tau\{Y_{ij}(s)\} = H(u_{ij})^\top \eta(s) + \xi_i^\top \{\phi_0(s) + u_{ij}\phi_1(s)\} + \zeta_{ij}^\top \psi(s).$$

The persistent component $B_{ij}(s) = \xi_i^\top \{\phi_0(s) + u_{ij}\phi_1(s)\}$ gives every subject its own functional intercept and slope in time, and the transient component $U_{ij}(s) = \zeta_{ij}^\top \psi(s)$ holds the deviations of single visits. The visit scores of a subject are orthogonal to its time design, $H_i^\top Z_i = 0$, where the rows of H_i and Z_i are $H(u_{ij})^\top$ and $\zeta_{ij}^\top$. Visit deviations thus carry no polynomial trend within a subject and, with centered subject scores, η is the average of the subjects' own trends.

`lfqpca` fits the model at one quantile level, and `lfqpca_continuation` reaches other levels from the median in small steps. The regression model

$$Q_{\tau_R}(V_{ij}) = G(u_{ij})^\top \alpha + C_{ij}^\top \beta + \langle \gamma_B, B_{ij} \rangle + \langle \gamma_U, U_{ij} \rangle$$

uses the components of an LFQPCA fit at level τ_F; it is fitted by `lfqpcr_from_lfqpca`, or together with the first stage by `lfqpcr`. The dimensions and the level τ_F can be chosen by `lfqpca_cv` and `lfqpcr_cv`, and `lfqpcr_bootstrap` resamples subjects.

The simulation designs are generated by `lfqpca_simulate` (Experiment I, designs D1 to D7) and `lfqpcr_simulate` (Experiment II, Case1 to Case6).

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`lfqpca` *Longitudinal functional quantile principal component analysis*

Description

Fits the LFQPCA model at one quantile level to curves recorded at several visits per subject.

Usage

```
lfqpca(Y, subject, visit_time, grid = NULL, tau = 0.5, K_subject = 2,
       K_visit = 1, spline_df = 8, spline_degree = 3, periodic = FALSE,
       mean_time_degree = 1, max_iter = 30, tol = 1e-4, n_starts = 2,
       rq_method_functions = c("fnb", "fn", "br"),
       rq_method_scores = c("br", "fnb", "fn"), strict_identification = TRUE,
       objective_increase_tolerance = 1e-7, verbose = TRUE,
       warm_start_fit = NULL, rank_tol = 1e-8)

## S3 method for class lfqpca_fit
print(x, ...)

## S3 method for class lfqpca_fit
summary(object, ...)

## S3 method for class summary.lfqpca_fit
print(x, ...)

## S3 method for class lfqpca_fit
fitted(object, ...)

## S3 method for class lfqpca_fit
residuals(object, ...)
```

Arguments

`Y`	numeric matrix with one curve per row and one column per grid point. Missing values are treated as unobserved; each curve needs at least five observed values.
`subject`	subject labels, one for each row of `Y`.
`visit_time`	visit times, one for each row of `Y`, for example years since the first visit.
`grid`	increasing grid of the functional domain. The default is an equally spaced grid on [0, 1].
`tau`	quantile level in (0, 1).
`K_subject`	number of subject components.
`K_visit`	number of visit components; 0 gives a model without visit deviations.
`spline_df`	dimension of the B-spline basis used for all functions.
`spline_degree`	degree of the B-splines; 3 gives cubic splines.
`periodic`	if `TRUE`, periodic B-splines are used, for curves on a circular domain.
`mean_time_degree`	degree p of the polynomial time basis.

`max_iter`	maximum number of iterations for each start.
`tol`	tolerance for the relative change of the objective.
`n_starts`	number of starting values; see Details.
`rq_method_functions`, `rq_method_scores`	quantile regression algorithms for the function and the score updates: `"fnb"` and `"fn"` both use the Frisch-Newton interior point method and `"br"` the simplex method of Barrodale and Roberts, both from **quantreg**. If the interior point method fails, the simplex method is tried.
`strict_identification`	if `TRUE` and `K_visit > 0`, every subject needs at least $p + 2$ distinct visit times. With `FALSE`, subjects with fewer visits are kept and get zero visit scores.
`objective_increase_tolerance`	largest increase of the objective accepted in a block update.
`verbose`	if `TRUE`, progress is reported by messages.
`warm_start_fit`	optional `lfqpca_fit` for the same data and dimensions, usually at a nearby quantile level; it is used as the second start.
`rank_tol`	tolerance for the rank checks of the fitted subject and visit blocks.
`x`, `object`	an object of class `lfqpca_fit`, or of class `summary.lfqpca_fit` for the print method of the summary.
`...`	not used.

Details

With u_{ij} the visit time standardized by its mean and standard deviation and $H(u) = (1, u, \ldots, u^p)^\top$, the fitted quantile curves are

$$Q_{ij}(s) = H(u_{ij})^\top \eta(s) + \xi_i^\top \{\phi_0(s) + u_{ij}\phi_1(s)\} + \zeta_{ij}^\top \psi(s).$$

The visit scores of subject i are $Z_i = N_i A_i$, where the columns of N_i are an orthonormal basis of the null space of $H_i^\top$, so that $H_i^\top Z_i = 0$. The subject scores have mean zero, the pairs (ϕ_{0k}, ϕ_{1k}) are orthonormal for $\int f_0 g_0 + \int f_1 g_1$, the ψ_l are orthonormal, and in each block the scores are uncorrelated with decreasing variances. Each component is signed so that its largest loading value is positive.

All functions lie in the span of the B-spline basis, and the estimate minimizes

$$L_\tau = \sum_{i=1}^{I} \sum_{j=1}^{J_i} \sum_{d \in O_{ij}} \frac{\omega_{ijd}}{I J_i} \rho_\tau\{Y_{ij}(s_d) - Q_{ij}(s_d)\},$$

where $\rho_\tau(r) = r\{\tau - 1(r < 0)\}$ is the check function, O_{ij} holds the observed grid points of curve (i, j) and the trapezoidal weights ω_{ijd} sum to one over these points. Every subject has the same total weight, whatever its number of visits or missing values.

The algorithm cycles through three blocks. Given the scores, the spline coefficients of all functions solve one weighted linear quantile regression. Given the functions, the scores (ξ_i, A_i) of each subject solve a small weighted quantile regression. A rotation then restores the identifying conditions without changing the fitted curves. An update is accepted only if it raises the objective by at most `objective_increase_tolerance`; the fit has converged when the relative change of the objective is below `tol` in an iteration without rejected updates.

The first start fits the population part alone and takes singular value decompositions of subject-wise intercept and slope curves of the residuals and of the remaining visit residuals. With a warm start, the second start is the fit given in `warm_start_fit`; the other starts are random. The fit with the smallest objective among the starts with full-rank subject and visit blocks is returned, or among all successful starts if none has full rank.

Value

`lfqpca` returns an object of class `lfqpca_fit`, a list with components including

`eta_functions` matrix of the population functions on the grid, one column per term of $H(u)$.

`phi0`, `phi1` matrices of the subject intercept and slope loadings, one column per component.

`psi` matrix of the visit loadings.

`xi` matrix of subject scores, one row per subject in order of first appearance in `subject`.

`zeta` matrix of visit scores, one row per curve.

`lambda`, `nu` variances of the subject and visit scores; `pve_subject` and `pve_visit` give their cumulative proportions.

`fitted` matrix of fitted quantile curves.

`objective`, `objective_history`
value of the criterion at the solution and over the iterations.

`converged`, `termination`, `iterations`
convergence flag, reason for stopping and number of iterations. The reason is one of `"converged"`, `"max_iter"` and `"stalled_rejected_updates"`.

`nondegenerate`, `rank_diagnostics`
whether the subject and visit blocks have full rank, and the rank checks behind it.

`constraint_diagnostics`
largest absolute entries of $H_i^\top Z_i$ for each subject.

`time_center`, `time_scale`
mean and standard deviation used to standardize `visit_time`.

`all_start_objectives`, `start_types`, `best_start`
objectives and types of the starts, and the index of the returned start.

The data, grid, weights, spline and time bases and the settings of the call are kept as well.

`summary` returns the quantile level, convergence information, the weighted proportion of observed values at or below the fitted quantiles, the score variances and their cumulative proportions, and the largest constraint errors. `fitted` and `residuals` return the fitted curves and `Y` minus the fitted curves.

Examples

```
## Not run:
# Experiment I, design D1, first replication
dat <- lfqpca_simulate("D1", I = 60, J = 5, replication = 1)
fit <- lfqpca(dat$Y, dat$subject, dat$visit_time, dat$grid, tau = 0.5,
              K_subject = 2, K_visit = 2, spline_df = 10,
              mean_time_degree = 2, max_iter = 60, tol = 1e-5, n_starts = 5,
              rq_method_functions = "fnb", rq_method_scores = "br",
              verbose = FALSE)
fit
summary(fit)

# Error of the fitted median surface
truth <- lfqpca_sim_truth(dat, tau = 0.5)
sqrt(mean((fitted(fit) - truth$quantile_surface)^2))

## End(Not run)
```

`lfqpca_continuation` *LFQPCA along a path of quantile levels*

Description

Fits LFQPCA at a target quantile level by moving from a reference level in steps, each fit starting from the previous one.

Usage

```
lfqpca_continuation(Y, subject, visit_time, grid = NULL, tau,
                    reference_tau = 0.5, fractions = c(0, 0.5, 0.75, 1),
                    n_starts = 5, path_starts = 2, reference_fit = NULL,
                    verbose = FALSE, ...)

lfqpca_continuation_path(tau, reference_tau = 0.5,
                         fractions = c(0, 0.5, 0.75, 1))
```

Arguments

`Y`, `subject`, `visit_time`, `grid`	the data, as in `lfqpca`.
`tau`	target quantile level.
`reference_tau`	level at which the path starts.
`fractions`	fractions of the distance from `reference_tau` to `tau` at which the model is fitted; 0 and 1 are always included.
`n_starts`	number of starts of the reference fit and of the fit at `tau`.
`path_starts`	number of starts of the intermediate fits; at least 2, since one of them is the warm start.
`reference_fit`	optional fit at `reference_tau`. It replaces the reference fit, which saves time when several levels share one reference.
`verbose`	passed to `lfqpca`.
`...`	further arguments of `lfqpca`, for example `K_subject`, `K_visit`, `spline_df`, `mean_time_degree`, `max_iter` or `tol`.

Details

The criterion is not convex in the functions and scores jointly, so the block algorithm finds a local minimum that depends on its start. The continuation fits the model at $\tau_0 + f(\tau - \tau_0)$ for each fraction f, where τ_0 is `reference_tau`, and passes each fit to the next one as `warm_start_fit`. The structured and the random starts of `lfqpca` stay available at every step. With the default fractions, a fit at 0.9 goes through 0.5, 0.7 and 0.8.

A failed fit at the reference level or at an intermediate level is recorded and skipped, and the next fit starts from the last successful one. Only a failure at `tau` is an error.

Value

The fit at `tau` is returned, with two further components: the messages of failed fits along the path, `continuation_errors`, and the levels of the path, `continuation_tau_path`. The path function returns these levels only.

Examples

```
## Not run:
# Levels visited on the way from 0.5 to 0.9
lfqpca_continuation_path(tau = 0.9)

# Experiment I, design D3, at tau = 0.9 as in the simulations
dat <- lfqpca_simulate("D3", I = 60, J = 5, replication = 1)
fit90 <- lfqpca_continuation(dat$Y, dat$subject, dat$visit_time, dat$grid,
                             tau = 0.9, K_subject = 2, K_visit = 2,
                             spline_df = 10, mean_time_degree = 2,
                             max_iter = 60, tol = 1e-5,
                             rq_method_functions = "fnb",
                             rq_method_scores = "br")
fit90$continuation_tau_path
fit90$continuation_errors
truth90 <- lfqpca_sim_truth(dat, tau = 0.9)
sqrt(mean((fitted(fit90) - truth90$quantile_surface)^2))

# Lower tail, starting from an existing median fit
fit50 <- lfqpca(dat$Y, dat$subject, dat$visit_time, dat$grid, tau = 0.5,
                K_subject = 2, K_visit = 2, spline_df = 10,
                mean_time_degree = 2, max_iter = 60, tol = 1e-5, n_starts = 5,
                rq_method_functions = "fnb", rq_method_scores = "br",
                verbose = FALSE)
fit10 <- lfqpca_continuation(dat$Y, dat$subject, dat$visit_time, dat$grid,
                             tau = 0.1, reference_fit = fit50,
                             K_subject = 2, K_visit = 2, spline_df = 10,
                             mean_time_degree = 2, max_iter = 60, tol = 1e-5,
                             rq_method_functions = "fnb",
                             rq_method_scores = "br")
fit10$continuation_tau_path

## End(Not run)
```

`lfqpca_cv` *Grid-point cross-validation for LFQPCA*

Description

Compares numbers of components and spline dimensions by holding out grid points within the curves.

Usage

```
lfqpca_cv(Y, subject, visit_time, grid = NULL,
          tuning_grid = expand.grid(K_subject = c(1, 2), K_visit = c(1, 2),
                                    spline_df = c(6, 8),
                                    KEEP.OUT.ATTRS = FALSE),
          tau = 0.5, n_folds = 3, max_iter = 20, n_starts = 1, verbose = TRUE,
          ...)
```

Arguments

Y, subject, visit_time, grid	the data, as in lfqpca.
tuning_grid	data frame with columns K_subject, K_visit and spline_df, one row per candidate model.
tau	quantile level.
n_folds	number of folds.
max_iter, n_starts	passed to lfqpca.
verbose	if TRUE, progress is reported by messages.
...	further arguments of lfqpca, for example mean_time_degree.

Details

The observed points of every curve are split at random into n_folds folds of nearly equal size. For each candidate and each fold, the model is fitted with the points of the fold removed, and the held-out loss is the check loss at these points with the same subject-balanced weights as the criterion of lfqpca. Since only grid points are held out, every subject and visit takes part in every fit. Each curve needs at least n_folds + 5 observed points. The folds are drawn with the current random number generator, so set.seed makes the result reproducible.

Value

A list with components

fold_results	data frame with the held-out loss of every candidate and fold.
summary	mean and standard deviation of the held-out loss of each candidate, ordered by the mean.
best	first row of summary.
fold_id	matrix with the fold of every observed grid point.

Examples

```
## Not run:
dat <- lfqpca_simulate("D1", I = 60, J = 5, replication = 1)
tuning <- expand.grid(K_subject = 1:3, K_visit = 1:2, spline_df = c(8, 10),
                      KEEP.OUT.ATTRS = FALSE)
set.seed(1)
cv <- lfqpca_cv(dat$Y, dat$subject, dat$visit_time, dat$grid,
                tuning_grid = tuning, tau = 0.5, n_folds = 3, max_iter = 20,
                verbose = FALSE, mean_time_degree = 2)
cv$summary
cv$best

## End(Not run)
```

`lfqpca_simulate` *Simulation designs of Experiment I*

Description

Generates data from the seven designs used to study LFQPCA in the paper.

Usage

```
lfqpca_simulate(design = paste0("D", 1:7), I = 60, J = 5, replication = 1,
                seed = NULL, missing_rate = 0.05,
                settings = lfqpca_sim_settings())

lfqpca_sim_data(design = paste0("D", 1:7), seed, I = 60, J = 5,
                missing_rate = 0.05, settings = lfqpca_sim_settings(),
                scale_only = FALSE)
```

Arguments

`design`	one of `"D1"` to `"D7"`; see Details.
`I`	number of subjects.
`J`	number of visits per subject; ignored in design D6.
`replication`	replication number, which sets the seed.
`seed`	seed of the data set. In `lfqpca_simulate` it overrides `replication`.
`missing_rate`	proportion of curve values removed at random.
`settings`	design constants, from `lfqpca_sim_settings`.
`scale_only`	if `TRUE`, only the scale surface is generated and its minimum is returned. This is used by the reserve-seed rule and by `lfqpca_sim_calibrate`.

Details

Every subject has `J` visits (in D6, a number drawn from 4 to 9), with times drawn uniformly on [-1.2, 1.2] and sorted. The curves are observed at 61 equally spaced points of [0, 1] and follow the location-scale model

$$Y_{ij}(s) = L_{ij}(s) + S_{ij}(s)\,\varepsilon_{ij}(s),$$

with independent standard normal errors, so that the τ-quantile is $L_{ij} + \Phi^{-1}(\tau)S_{ij}$. The location L has the LFQPCA form with a quadratic time basis, two subject components and two visit components, orthonormal trigonometric loadings, centered normal subject scores with standard deviations 1.25 and 0.70, and visit scores $N_i A_i$ with standard deviations 0.75 and 0.40. The population functions are $1.2 + 0.8\sin(2\pi s) + 0.2\cos(4\pi s)$ for the intercept, $0.35\cos(2\pi s) + 0.2(s - 0.5)$ for u and $-0.18\sin(\pi s)$ for u^2.

D1 time effects, with subject and visit components common to all quantile levels; $S(s) = 0.75 + 0.15\cos(2\pi s)$.

D2 as D1 without time effects: the population functions of u and u^2 and the slope loadings are zero.

D3 quantile-specific scores. The scale has its own first subject and visit scores on the location loadings, so the scores at level τ are $\xi_i + \Phi^{-1}(\tau)\xi_i^S$, and similarly for the visit scores. The first location visit score is doubled, and the scale intercept keeps S above `min_scale` when $|u| \leq 2.2$.

D4 quantile-specific loadings. The scale uses the location scores with new loading directions orthogonal to the location loadings, and the intercept 3 from `lfqpca_sim_calibrate`.

D5 visit scores drawn independently, which violates $H_i^\top Z_i = 0$.

D6 unbalanced visits.

D7 subject curvature outside the model: the term $\xi_i^\top \phi_2(s) u_{ij}^2$ with $\phi_2 = 1.13\,\phi_1$ is added.

`lfqpca_simulate` uses the seed `seed_experiment1 + replication` for every design. In D3 and D4, a draw whose scale surface falls below `min_scale` is replaced by the draw with seed `seed + k * reserve_seed_offset` for the smallest k that gives an admissible surface, with k at most `reserve_seeds`; `reserve_draws` records k. `lfqpca_sim_data` generates one data set from a given seed, without this rule.

Value

A list with components `design`, `seed`, `Y` (curves with missing values), `Y_complete`, `subject`, `visit_time`, `time_std`, `grid`, `weights` (trapezoidal weights), `visits` (visits per subject) and `truth` (generating functions, scores and surfaces of the location and scale parts). `lfqpca_simulate` adds `reserve_draws`. With `scale_only = TRUE`, a short list with the scale intercept and the minimum of the scale surface is returned.

Examples

```
## Not run:
# Design D3, replication 1
dat <- lfqpca_simulate("D3", I = 60, J = 5, replication = 1)
dim(dat$Y)
mean(is.na(dat$Y))

# One replication of each design
sims <- lapply(paste0("D", 1:7), lfqpca_simulate, I = 60, J = 5,
               replication = 1)
names(sims) <- paste0("D", 1:7)
sapply(sims, function(x) nrow(x$Y))

# Unbalanced visits, and the same data from lfqpca_sim_data
dat6 <- lfqpca_simulate("D6", I = 120, replication = 3)
table(dat6$visits)
seed3 <- lfqpca_sim_settings()$seed_experiment1 + 3
dat6b <- lfqpca_sim_data("D6", seed = seed3, I = 120)
identical(dat6$Y, dat6b$Y)

## End(Not run)
```

`lfqpca_sim_calibrate` *Calibration of the D4 scale intercept*

Description

Pilot computation of the intercept of the scale surface in design D4.

Usage

```
lfqpca_sim_calibrate(n_rep = 500, I = 120, J = c(5, 10), start = 0.75,
                     margin = 0.25, rounding = 0.1, seed = NULL,
                     settings = lfqpca_sim_settings())
```

Arguments

n_rep	number of probe draws for each number of visits.
I	number of subjects in the probe draws.
J	numbers of visits; one pilot is run for each.
start	intercept used in the probe draws.
margin	amount added to the largest required intercept.
rounding	the result is rounded up to a multiple of rounding.
seed	base seed of the probe draws; by default seed_experiment1 + 1e6.
settings	design constants, from lfqpca_sim_settings.

Details

For each probe draw, the intercept needed to keep the D4 scale surface above min_scale is start plus the shortfall of the surface. The value for each J is the largest of these intercepts plus margin, rounded up, and the result is the largest value over J. With the defaults the largest required intercepts are 2.677 for five visits and 2.394 for ten visits, which gives the value 3 used in the paper. The reserve-seed rule of lfqpca_simulate covers the rare draws that still fall below the bound.

Value

A list with the calibrated intercept and a data frame by_J with the largest required intercept and the rounded value for each J.

Examples

```
## Not run:
cal <- lfqpca_sim_calibrate(n_rep = 500, I = 120, J = c(5, 10))
cal$by_J
cal$intercept

## End(Not run)
```

---

lfqpca_sim_settings *Constants and list of the simulation designs*

Description

lfqpca_sim_settings returns the constants of the simulation designs and lfqpca_sim_designs lists the designs of both experiments.

Usage

```
lfqpca_sim_settings()

lfqpca_sim_designs()
```

Details

The main constants are the number of grid points `D` (61), the numbers of subject and visit components (2 each), the degree of the time basis (2), the spline dimension used in the fits (10), the range of the visit times (`time_range`), the numbers of visits in the unbalanced designs (`unbalanced_visits`, 4 to 9), the scale intercepts of the designs with a common scale and of D4 (`base_scale_intercept`, `loading_scale_intercept`) and the lower bound `min_scale` of the scale surfaces.

In D3 and Case 3, the scale scores are chosen so that the first subject scores at the reference and the calibration levels (`reference_tau`, `calibration_tau`) have the correlation `score_correlation`; the first visit score is multiplied by `visit_signal`, and the scale intercept keeps the scale above `min_scale` for $|u| \leq$ `max_abs_time` and at most `bound_visits` visits. `loading_norms` are the norms of the D4 scale loadings and `curvature` is the factor of the D7 quadratic term. In Case 3 the response depends on the components at the level `response_tau_star`. The seeds of the two experiments are `seed_experiment1` and `seed_experiment2`; `iid_seed_offset`, `reserve_seed_offset` and `reserve_seeds` set the seeds of the independent visit scores and of the reserve draws.

Value

`lfqpca_sim_settings` returns a named list. `lfqpca_sim_designs` returns a data frame with columns `experiment`, `code` and `description`.

Examples

```
## Not run:
lfqpca_sim_designs()
settings <- lfqpca_sim_settings()
str(settings[c("D", "K_subject", "K_visit", "spline_df", "min_scale")])

# Curves on a coarser grid
settings$D <- 31
dat <- lfqpca_simulate("D1", I = 30, settings = settings)
dim(dat$Y)

## End(Not run)
```

---

lfqpca_sim_truth *True components of an Experiment I data set*

Description

Returns the true population surface, components and quantile curves of a simulated data set at a given quantile level.

Usage

```
lfqpca_sim_truth(data, tau, settings = lfqpca_sim_settings())
```

Arguments

`data`	output of `lfqpca_simulate` or `lfqpca_sim_data`.
`tau`	quantile level.
`settings`	design constants, from `lfqpca_sim_settings`.

Details

The τ-quantile is the location plus $q = \Phi^{-1}(\tau) - \Phi^{-1}(0.5)$ times the scale, split into population, subject and visit parts. In D3 the scores move with q, in D4 the loadings do, and in the other designs only the population part changes. In D7, `B` contains the quadratic subject term and `B_intercept_slope` does not. The true functions need not satisfy the normalization of `lfqpca`, so a fit is best compared with the surfaces `population`, `B`, `U` and `quantile_surface`, or through the spans of the loadings.

Value

A list with the level `tau`, the shift `q_shift`, the functions `eta`, `phi0`, `phi1` and `psi`, the scores `xi` and `zeta`, and the surfaces `population`, `B_intercept_slope`, `B`, `U` and `quantile_surface`, which have one row per curve and one column per grid point.

Examples

```
## Not run:
dat <- lfqpca_simulate("D3", I = 60, J = 5, replication = 1)
truth <- lfqpca_sim_truth(dat, tau = 0.9)
matplot(dat$grid, t(truth$quantile_surface[dat$subject == 1, ]), type = "l",
        lty = 1, xlab = "s", ylab = "0.9-quantile, subject 1")

# Share of the subject part that is due to the quadratic term in D7
dat7 <- lfqpca_simulate("D7", I = 60, J = 5, replication = 1)
truth7 <- lfqpca_sim_truth(dat7, tau = 0.5)
sum((truth7$B - truth7$B_intercept_slope)^2) / sum(truth7$B^2)

## End(Not run)
```

`lfqpcr` *Longitudinal functional quantile principal component regression*

Description

Fits LFQPCA to the curves at level τ_F and then the quantile regression of a scalar response at level τ_R on the persistent and transient components.

Usage

```
lfqpcr(Y, V, subject, visit_time, grid = NULL, covariates = NULL, tau_F = 0.5,
       tau_R = 0.5, K_subject = 2, K_visit = 1, spline_df = 8,
       spline_degree = 3, periodic = FALSE, mean_time_degree = 1,
       response_time_degree = NULL, lfqpca_max_iter = 30, lfqpca_tol = 1e-4,
       lfqpca_n_starts = 2, lfqpca_rq_method_functions = c("fnb", "fn", "br"),
       lfqpca_rq_method_scores = c("br", "fnb", "fn"),
       strict_identification = TRUE, objective_increase_tolerance = 1e-7,
       lfqpcr_rq_method = c("fnb", "br", "fn"), space_tol = 1e-10,
       rank_tol = 1e-7, verbose = TRUE)

## S3 method for class lfqpcr_fit
print(x, ...)
```

```
## S3 method for class lfqpcr_fit
summary(object, ...)

## S3 method for class summary.lfqpcr_fit
print(x, ...)

## S3 method for class lfqpcr_fit
coef(object, ...)

## S3 method for class lfqpcr_fit
fitted(object, ...)

## S3 method for class lfqpcr_fit
residuals(object, ...)
```

Arguments

`Y`, `subject`, `visit_time`, `grid`	the curves and their design, as in `lfqpca`.
`V`	scalar responses, one for each row of `Y`; missing values are allowed.
`covariates`	optional numeric matrix or data frame of scalar covariates, one row for each row of `Y`.
`tau_F`	quantile level of the curves.
`tau_R`	quantile level of the response.
`K_subject`, `K_visit`, `spline_df`, `spline_degree`, `periodic`, `mean_time_degree`	dimensions and bases of the first stage, as in `lfqpca`.
`response_time_degree`, `space_tol`, `rank_tol`	as in `lfqpcr_from_lfqpca`.
`lfqpca_max_iter`, `lfqpca_tol`, `lfqpca_n_starts`	`max_iter`, `tol` and `n_starts` of the first stage.
`lfqpca_rq_method_functions`, `lfqpca_rq_method_scores`	quantile regression algorithms of the first stage.
`strict_identification`, `objective_increase_tolerance`	as in `lfqpca`.
`lfqpcr_rq_method`	quantile regression algorithm of the second stage.
`verbose`	if `TRUE`, progress of the first stage is reported.
`x`, `object`	an object of class `lfqpcr_fit`, or of class `summary.lfqpcr_fit` for the print method of the summary.
`...`	not used.

Details

The first stage calls `lfqpca` directly, without quantile continuation. For levels τ_F far from the median it can be better to fit the first stage with `lfqpca_continuation` and pass the result to `lfqpcr_from_lfqpca`, which is how the simulations of the paper were run. The model and the estimator of the second stage are described with `lfqpcr_from_lfqpca`.

Value

An object of class `lfqpcr_fit`, as returned by `lfqpcr_from_lfqpca`. `coef`, `fitted` and `residuals` return the regression coefficients, the fitted τ_R-quantiles and the residuals of `V`. `summary` returns the quantile levels, the number of subjects with a response, the dimensions of the loading spaces, the rank of the design, the number of observations, the criterion, the subject-balanced proportion of responses at or below the fitted quantiles and the coefficients.

Examples

```
## Not run:
# Experiment II, Case 2, with tau_F = 0.5 and tau_R = 0.9
sim <- lfqpcr_simulate("Case2", I_train = 60, I_test = 20, J = 5,
                       tau_F = 0.5, tau_R = 0.9, replication = 1)
tr <- sim$train
te <- sim$test
fit <- lfqpcr(sim$Y[tr, ], sim$V[tr], sim$subject[tr], sim$visit_time[tr],
              sim$grid, covariates = sim$C[tr, ], tau_F = 0.5, tau_R = 0.9,
              K_subject = 2, K_visit = 2, spline_df = 10,
              mean_time_degree = 2, response_time_degree = 2,
              lfqpca_max_iter = 60, lfqpca_tol = 1e-5, lfqpca_n_starts = 5,
              lfqpca_rq_method_functions = "fnb",
              lfqpca_rq_method_scores = "br", verbose = FALSE)
fit
summary(fit)
head(fitted(fit))

## End(Not run)
```

---

lfqpcr_bootstrap *Subject bootstrap for LFQPCR*

Description

Refits LFQPCR to samples of subjects drawn with replacement.

Usage

```
lfqpcr_bootstrap(Y, V, subject, visit_time, grid = NULL, covariates = NULL,
                 B = 200, seed = 1, ..., verbose = TRUE)
```

Arguments

`Y`, `V`, `subject`, `visit_time`, `grid`, `covariates`	the data, as in `lfqpcr`.
`B`	number of bootstrap samples.
`seed`	seed for the resampling. The random number state of the session is restored afterwards.
`...`	further arguments of `lfqpcr`, such as `tau_F`, `tau_R` and the dimensions.
`verbose`	if `TRUE`, progress is reported by messages.

Details

Each bootstrap sample draws as many subjects as the data contain, with replacement, and keeps all their visits; a subject drawn twice enters as two subjects. Fits that fail are kept as error objects. Percentile intervals for the coefficient functions or other quantities follow from the successful fits. A fit counts as successful when `lfqpcr` returns; the components `lfqpca_converged` and `lfqpca_nondegenerate` of each fit show whether its first stage converged with full-rank blocks.

Value

A list with components `fits` (list of `lfqpcr_fit` objects and error objects), `successful` (logical vector), `n_successful`, `n_failed` and `seed`.

Examples

```
## Not run:
sim <- lfqpcr_simulate("Case2", I_train = 60, I_test = 20, J = 5,
                       tau_F = 0.5, tau_R = 0.9, replication = 1)
tr <- sim$train
boot <- lfqpcr_bootstrap(sim$Y[tr, ], sim$V[tr], sim$subject[tr],
                         sim$visit_time[tr], sim$grid,
                         covariates = sim$C[tr, ], B = 50, seed = 1,
                         tau_F = 0.5, tau_R = 0.9, K_subject = 2,
                         K_visit = 2, spline_df = 10, mean_time_degree = 2,
                         response_time_degree = 2, lfqpca_max_iter = 60,
                         lfqpca_tol = 1e-5, lfqpca_n_starts = 5)
boot$n_successful

# Pointwise 95% percentile band for gamma_B
gamma_B <- sapply(boot$fits[boot$successful], function(f) f$gamma_B)
band <- apply(gamma_B, 1, quantile, probs = c(0.025, 0.975))
matplot(sim$grid, cbind(t(band), sim$truth$gamma_B), type = "l",
        lty = c(2, 2, 1), col = 1, xlab = "s", ylab = "gamma_B")

## End(Not run)
```

---

lfqpcr_coefficient_functions

Coefficient functions and time effect of an LFQPCR fit

Description

`lfqpcr_coefficient_functions` returns the coefficient functions γ_B and γ_U on the grid, and `lfqpcr_alpha` evaluates the time effect $G(u)^\top \alpha$ at given visit times.

Usage

```
lfqpcr_coefficient_functions(object)

lfqpcr_alpha(object, visit_time)
```

Arguments

object	an object of class lfqpcr_fit.
visit_time	visit times on the scale of the original visit_time.

Details

If the persistent component of a visit changes by a function h, the fitted τ_R-quantile of the response changes by $\int \gamma_B h$; γ_U plays the same role for the transient component. Both functions lie in the spans of the fitted loadings.

Value

lfqpcr_coefficient_functions returns a data frame with columns grid, gamma_B and gamma_U. lfqpcr_alpha returns a numeric vector with one value per visit time.

Examples

```
## Not run:
sim <- lfqpcr_simulate("Case2", I_train = 60, I_test = 20, J = 5,
                       tau_F = 0.5, tau_R = 0.9, replication = 1)
tr <- sim$train
te <- sim$test
fit <- lfqpcr(sim$Y[tr, ], sim$V[tr], sim$subject[tr], sim$visit_time[tr],
              sim$grid, covariates = sim$C[tr, ], tau_F = 0.5, tau_R = 0.9,
              K_subject = 2, K_visit = 2, spline_df = 10,
              mean_time_degree = 2, response_time_degree = 2,
              lfqpca_max_iter = 60, lfqpca_tol = 1e-5, lfqpca_n_starts = 5,
              lfqpca_rq_method_functions = "fnb",
              lfqpca_rq_method_scores = "br", verbose = FALSE)
gam <- lfqpcr_coefficient_functions(fit)
matplot(gam$grid, cbind(gam$gamma_B, sim$truth$gamma_B), type = "l",
        lty = 1:2, col = 1, xlab = "s", ylab = "gamma_B")
lfqpcr_alpha(fit, visit_time = c(-1, 0, 1))

## End(Not run)
```

---

lfqpcr_cv *Subject-level cross-validation for LFQPCR*

Description

Estimates the prediction loss of LFQPCR by leaving out groups of subjects.

Usage

```
lfqpcr_cv(Y, V, subject, visit_time, grid = NULL, covariates = NULL,
          tau_F = 0.5, tau_R = 0.5, K_subject = 2, K_visit = 1, spline_df = 8,
          spline_degree = 3, periodic = FALSE, mean_time_degree = 1,
          response_time_degree = NULL, n_folds = 5, seed = 1,
          lfqpca_max_iter = 30, lfqpca_tol = 1e-4, lfqpca_n_starts = 2,
          lfqpca_rq_method_functions = c("fnb", "fn", "br"),
          lfqpca_rq_method_scores = c("br", "fnb", "fn"),
```

```
            strict_identification = TRUE, objective_increase_tolerance = 1e-7,
            lfqpcr_rq_method = c("fnb", "br", "fn"), space_tol = 1e-10,
            rank_tol = 1e-7, verbose = TRUE)
```

Arguments

Y, V, subject, visit_time, grid, covariates	the data, as in lfqpcr.
tau_F, tau_R, K_subject, K_visit, spline_df, spline_degree, periodic	settings of lfqpcr.
mean_time_degree, response_time_degree, strict_identification	settings of lfqpcr.
lfqpca_max_iter, lfqpca_tol, lfqpca_n_starts	settings of the first stage.
lfqpca_rq_method_functions, lfqpca_rq_method_scores, lfqpcr_rq_method	quantile regression algorithms of the two stages.
objective_increase_tolerance, space_tol, rank_tol	tolerances, as in lfqpcr.
n_folds	number of folds, between 2 and the number of subjects.
seed	seed for the allocation of subjects to folds. The random number state of the session is restored afterwards.
verbose	if TRUE, progress is reported by messages.

Details

The subjects are allocated at random to n_folds folds. For each fold, lfqpcr is fitted to the other subjects, and the responses of the held-out subjects are predicted from their curves with the predict method. The criterion cv_objective is the subject-balanced check loss at level tau_R over all held-out predictions. Comparing it over tau_F, the numbers of components or spline_df gives a choice of these settings.

Value

A list with components tau_F, tau_R, n_folds, fold_id (fold of each row), prediction (held-out predictions), eligible (rows used in the criterion), response_weights and cv_objective.

Examples

```
## Not run:
# Choice of tau_F by five-fold cross-validation in Case 3
sim <- lfqpcr_simulate("Case3", I_train = 60, I_test = 20, J = 5,
                       tau_F = 0.9, tau_R = 0.9, replication = 1)
tr <- sim$train
tau_F_grid <- c(0.1, 0.5, 0.9)
cv_loss <- sapply(tau_F_grid, function(tf) {
  lfqpcr_cv(sim$Y[tr, ], sim$V[tr], sim$subject[tr], sim$visit_time[tr],
            sim$grid, covariates = sim$C[tr, ], tau_F = tf, tau_R = 0.9,
            K_subject = 2, K_visit = 2, spline_df = 10, mean_time_degree = 2,
            response_time_degree = 2, n_folds = 5, seed = 1,
            lfqpca_max_iter = 60, lfqpca_tol = 1e-5, lfqpca_n_starts = 5,
            verbose = FALSE)$cv_objective
})
```

```
tau_F_grid[which.min(cv_loss)]

## End(Not run)
```

`lfqpcr_from_lfqpca` *Quantile regression on the persistent and transient components*

Description

Second stage of LFQPCR: regresses a quantile of a scalar response on the subject and visit components of a fitted LFQPCA model.

Usage

```
lfqpcr_from_lfqpca(lfqpca_fit, V, covariates = NULL, tau_R = 0.5,
                   response_time_degree = NULL,
                   rq_method = c("fnb", "br", "fn"), space_tol = 1e-10,
                   rank_tol = 1e-7)
```

Arguments

`lfqpca_fit`	an object of class `lfqpca_fit`, fitted at level τ_F.
`V`	scalar responses, one for each curve of the fit. Visits with a missing response are left out.
`covariates`	optional numeric matrix or data frame of scalar covariates, one row for each curve of the fit.
`tau_R`	quantile level of the response.
`response_time_degree`	degree of the time basis $G(u)$; at least `mean_time_degree` of the fit, which is the default.
`rq_method`	quantile regression algorithm, as in `lfqpca`.
`space_tol`	relative tolerance on the singular values when the bases of the loading spaces are built.
`rank_tol`	tolerance for removing linearly dependent columns of the design.

Details

The model for the τ_R-quantile of the response is

$$Q_{\tau_R}(V_{ij}) = G(u_{ij})^\top \alpha + C_{ij}^\top \beta + \langle \gamma_B, B_{ij} \rangle + \langle \gamma_U, U_{ij} \rangle,$$

where B_{ij} and U_{ij} are the persistent and transient components of the LFQPCA fit and the time basis is $G(u) = (1, u, \ldots, u^{p_G})^\top$. Inner products $\langle f, g \rangle = \int fg$ are computed with the trapezoidal rule on the grid. Only the parts of γ_B in the span of the loadings ϕ_{0k}, ϕ_{1k} and of γ_U in the span of the ψ_l enter the model, so the coefficient functions are taken in these spaces.

Orthonormal bases χ_B and χ_U of the two spaces come from singular value decompositions of the weighted loadings. With the coordinates $x_{B,ij} = \langle \chi_B, B_{ij} \rangle$ and $x_{U,ij} = \langle \chi_U, U_{ij} \rangle$, the estimate minimizes

$$\frac{1}{I_R} \sum_i \frac{1}{J_i^R} \sum_{j \in R_i} \rho_{\tau_R} \left(V_{ij} - G(u_{ij})^\top \alpha - C_{ij}^\top \beta - x_{B,ij}^\top a_B - x_{U,ij}^\top a_U \right),$$

where R_i holds the J_i^R visits of subject i with an observed response and I_R is the number of such subjects. Then $\gamma_B = a_B^\top \chi_B$ and $\gamma_U = a_U^\top \chi_U$. Linearly dependent columns of the design are removed before the fit, and their coefficients are set to zero.

In terms of the scores, $\langle \gamma_B, B_{ij} \rangle = \sum_k \xi_{ik}(\theta_{0k} + u_{ij}\theta_{1k})$ and $\langle \gamma_U, U_{ij} \rangle = \sum_l \zeta_{ijl}\delta_l$, with $\theta_{0k} = \langle \gamma_B, \phi_{0k} \rangle$, $\theta_{1k} = \langle \gamma_B, \phi_{1k} \rangle$ and $\delta_l = \langle \gamma_U, \psi_l \rangle$.

Value

An object of class `lfqpcr_fit`, a list with components including

`tau_F`, `tau_R` quantile levels of the curves and of the response.

`coefficients` named vector of the regression coefficients: time terms, covariates and coordinates.

`a_alpha`, `beta`, `a_B`, `a_U` the blocks of `coefficients`.

`gamma_B`, `gamma_U` coefficient functions on the grid.

`theta0`, `theta1`, `delta` coefficients of the score form.

`fitted`, `residuals` fitted τ_R-quantiles and residuals of `V`.

`objective` value of the criterion.

`B_hat`, `U_hat`, `x_B`, `x_U` fitted components and their coordinates.

`chi_B`, `chi_U`, `r_B`, `r_U` bases of the loading spaces and their dimensions.

`design_rank`, `dropped_columns` rank of the design and the removed columns.

`I_R`, `J_R`, `response_weights` number of subjects with a response, their numbers of visits and the weights of the criterion.

`lfqpca_fit`, `lfqpca_converged`, `lfqpca_nondegenerate` the first-stage fit and its convergence and full-rank flags. The regression is only as reliable as this fit.

Examples

```
## Not run:
# Experiment II, Case 2, with tau_F = tau_R = 0.9
sim <- lfqpcr_simulate("Case2", I_train = 60, I_test = 20, J = 5,
                       tau_F = 0.9, tau_R = 0.9, replication = 1)
tr <- sim$train

# First stage at tau_F = 0.9, reached from the median
fit_F <- lfqpca_continuation(sim$Y[tr, ], sim$subject[tr], sim$visit_time[tr],
                             sim$grid, tau = 0.9, K_subject = 2, K_visit = 2,
                             spline_df = 10, mean_time_degree = 2,
                             max_iter = 60, tol = 1e-5,
                             rq_method_functions = "fnb",
                             rq_method_scores = "br")

# Second stage at tau_R = 0.9
fit_R <- lfqpcr_from_lfqpca(fit_F, V = sim$V[tr], covariates = sim$C[tr, ],
```

```
                            tau_R = 0.9, response_time_degree = 2,
                            rq_method = "fnb")
fit_R
coef(fit_R)
matplot(sim$grid, cbind(fit_R$gamma_B, sim$truth$gamma_B), type = "l",
        lty = 1:2, col = 1, xlab = "s", ylab = "gamma_B")

## End(Not run)
```

`lfqpcr_simulate` *Simulation designs of Experiment II*

Description

Generates curves, scalar responses and covariates from the six cases used to study LFQPCR in the paper.

Usage

```
lfqpcr_simulate(case = paste0("Case", 1:6), I_train = 60, I_test = 20, J = 5,
                tau_F = 0.5, tau_R = 0.9, replication = 1, seed = NULL,
                settings = lfqpca_sim_settings())

lfqpcr_sim_data(case = paste0("Case", 1:6), seed, I_train = 60, I_test = 20,
                J = 5, tau_F = 0.5, tau_R = 0.9,
                settings = lfqpca_sim_settings())
```

Arguments

`case`	one of `"Case1"` to `"Case6"`; see Details.
`I_train`, `I_test`	numbers of training and test subjects.
`J`	number of visits per subject; ignored in Case 6.
`tau_F`	quantile level of the curves at which the true components are returned.
`tau_R`	quantile level of the response.
`replication`	replication number, which sets the seed.
`seed`	seed of the data set. In `lfqpcr_simulate` it overrides `replication`.
`settings`	design constants, from `lfqpca_sim_settings`.

Details

Visit times and grid are as in Experiment I, the curves are complete, and time is standardized with the visits of the training subjects. The location part of the curves has the form used in Experiment I, with subject scores centered over the training subjects. Except in Case 3, the curves are

$$Y_{ij}(s) = L_{ij}(s) + \sigma_{ij}(s)\{E_{ij}(s) - 1\}, \qquad \sigma_{ij}(s) = (0.55 + 0.2u_{ij}^2)(0.25 + 0.45s),$$

with standard exponential errors, so the persistent and transient components are the same at every level τ_F. The response is

$$V_{ij} = \alpha(u_{ij}) + C_{ij}^\top \beta + \langle \gamma_B, B_{ij} \rangle + \langle \gamma_U, U_{ij} \rangle + \sigma_R(u_{ij})\{E_{ij} - q_E(\tau_R)\},$$

with $\alpha(u) = 0.5 + 0.4u - 0.15u^2$, a subject-level and a visit-level standard normal covariate with $\beta = (0.5, -0.35)$, $\sigma_R(u) = 0.55 + 0.15|u|$, a standard exponential error E_{ij} and its quantile function q_E. The τ_R-quantile of V_{ij} is thus the linear predictor.

Case1 one coefficient function acting on $B + U$; its projections on the two loading spaces give γ_B and γ_U.

Case2 separate coefficient functions for B and U.

Case3 normal errors with quantile-specific scores as in design D3; the components change with τ_F, and the response depends on the components at level 0.9.

Case4 no time effect in the curves, the error scales or α.

Case5 visit scores drawn independently, which violates $H_i^\top Z_i = 0$.

Case6 unbalanced visits, 4 to 9 per subject.

Cases 2 to 6 share the coefficient functions. `lfqpcr_simulate` uses the seed `seed_experiment2 + replication - 1`; the curves and responses do not depend on `tau_F`.

Value

A list with components `case`, `seed`, `Y`, `V`, `C` (covariates), `subject`, `visit_time`, `grid`, `visits`, `train` and `test` (logical indicators of the rows), `tau_F`, `tau_R` and `truth`. The list `truth` holds the standardization of time, the true functions and scores at `tau_F`, the true τ_F-quantile curves (`functional_quantile`), the components that enter the response (`B_response`, `U_response`), the true `alpha`, `beta`, `gamma_B` and `gamma_U`, and the true response quantiles.

Examples

```
## Not run:
# Case 2, replication 1
sim <- lfqpcr_simulate("Case2", I_train = 60, I_test = 20, J = 5,
                       tau_F = 0.5, tau_R = 0.9, replication = 1)
table(train = sim$train)
str(sim$truth[c("beta", "gamma_B", "gamma_U")])

# One replication of each case at tau_F = 0.9
sims <- lapply(paste0("Case", 1:6), lfqpcr_simulate, tau_F = 0.9,
               tau_R = 0.9, replication = 1)
names(sims) <- paste0("Case", 1:6)
sapply(sims, function(x) length(x$V))

## End(Not run)
```

`plot.lfqpca_fit` *Plots of a fitted LFQPCA model*

Description

Draws the loadings, the objective over the iterations, observed and fitted curves, or population quantile curves of a fitted LFQPCA model.

Usage

```
## S3 method for class lfqpca_fit
plot(x,
  type = c("components", "objective", "fitted", "mean_surface"),
  curve_indices = NULL, mean_times = c(-1, 0, 1), ...)
```

Arguments

x	an object of class lfqpca_fit.
type	what to draw; see Details.
curve_indices	rows of Y drawn for type = "fitted"; by default the first six.
mean_times	standardized visit times at which the population curves are drawn for type = "mean_surface"; 0 is the mean visit time.
...	not used.

Details

"components" the subject intercept, subject slope and visit loadings;

"objective" the objective over the iterations;

"fitted" observed and fitted curves of the rows in curve_indices;

"mean_surface" population quantile curves at the standardized times mean_times.

Value

x, invisibly.

Examples

```
## Not run:
dat <- lfqpca_simulate("D1", I = 60, J = 5, replication = 1)
fit <- lfqpca(dat$Y, dat$subject, dat$visit_time, dat$grid, tau = 0.5,
              K_subject = 2, K_visit = 2, spline_df = 10,
              mean_time_degree = 2, max_iter = 60, tol = 1e-5, n_starts = 5,
              rq_method_functions = "fnb", rq_method_scores = "br",
              verbose = FALSE)
plot(fit, type = "components")
plot(fit, type = "objective")
plot(fit, type = "fitted", curve_indices = 1:4)
plot(fit, type = "mean_surface")

## End(Not run)
```

---

predict.lfqpca_fit *Scores and quantile curves of new subjects*

Description

Estimates the scores of new subjects from their curves, with the functions of a fitted LFQPCA model held fixed, and returns the fitted quantile curves.

Usage

```
## S3 method for class lfqpca_fit
predict(object, newY, new_subject, new_visit_time, ...)
```

Arguments

object	an object of class lfqpca_fit.
newY	matrix of curves of the new subjects on the grid of the fit, one per row; missing values are allowed.
new_subject	subject labels of the rows of newY.
new_visit_time	visit times of the rows of newY, on the scale of the original visit_time.
...	not used.

Details

The visit times are standardized with the mean and standard deviation of the training times. For each new subject, the scores (ξ, A) minimize the weighted check loss of its curves with $\eta, \phi_0, \phi_1, \psi$ fixed, and the visit scores are $N_0 A$, with N_0 a basis of the null space of the transposed time design of the subject. Each reconstructed curve therefore uses all curves of its subject. When the model has visit components, every new subject needs at least $p + 2$ distinct visit times.

Value

A list with components fitted (matrix of quantile curves), xi (subject scores, one row per new subject), zeta (visit scores), subject_id, subject_levels, time_std (standardized visit times), objective (criterion of the new curves) and constraint_diagnostics.

Examples

```
## Not run:
dat <- lfqpca_simulate("D1", I = 60, J = 5, replication = 1)
fit <- lfqpca(dat$Y, dat$subject, dat$visit_time, dat$grid, tau = 0.5,
              K_subject = 2, K_visit = 2, spline_df = 10,
              mean_time_degree = 2, max_iter = 60, tol = 1e-5, n_starts = 5,
              rq_method_functions = "fnb", rq_method_scores = "br",
              verbose = FALSE)

# Twenty new subjects from the same design
new <- lfqpca_simulate("D1", I = 20, J = 5, replication = 2)
pred <- predict(fit, newY = new$Y, new_subject = new$subject,
                new_visit_time = new$visit_time)
dim(pred$fitted)
head(pred$xi)

## End(Not run)
```

---

predict.lfqpcr_fit *Response quantiles of new subjects*

Description

Predicts the τ_R-quantile of the response of new subjects from their curves, times and covariates.

Usage

```
## S3 method for class lfqpcr_fit
predict(object, newY, new_subject, new_visit_time,
  new_covariates = NULL, trajectory_only = FALSE, ...)
```

Arguments

object	an object of class lfqpcr_fit.
newY, new_subject, new_visit_time	curves, subject labels and visit times of the new subjects, as in the predict method for lfqpca_fit objects.
new_covariates	covariates of the rows of newY, with the same columns as in the fit.
trajectory_only	if TRUE, the coordinates of the transient component are set to zero.
...	not used.

Details

The LFQPCA scores of the new subjects are estimated from their curves with the functions of the first stage held fixed. The persistent and transient components are rebuilt from these scores, and their coordinates enter the fitted regression. With trajectory_only = TRUE only the persistent component is used, so the prediction depends on the curves through the subject trajectory alone.

Value

A list with components fitted (predicted quantiles), x_B, x_U, B_hat, U_hat (coordinates and components), xi, zeta, time_std, subject_id and trajectory_only.

Examples

```
## Not run:
sim <- lfqpcr_simulate("Case2", I_train = 60, I_test = 20, J = 5,
                       tau_F = 0.5, tau_R = 0.9, replication = 1)
tr <- sim$train
te <- sim$test
fit <- lfqpcr(sim$Y[tr, ], sim$V[tr], sim$subject[tr], sim$visit_time[tr],
              sim$grid, covariates = sim$C[tr, ], tau_F = 0.5, tau_R = 0.9,
              K_subject = 2, K_visit = 2, spline_df = 10,
              mean_time_degree = 2, response_time_degree = 2,
              lfqpca_max_iter = 60, lfqpca_tol = 1e-5, lfqpca_n_starts = 5,
              lfqpca_rq_method_functions = "fnb",
              lfqpca_rq_method_scores = "br", verbose = FALSE)

# Prediction for the 20 test subjects and empirical coverage
pred <- predict(fit, newY = sim$Y[te, ], new_subject = sim$subject[te],
                new_visit_time = sim$visit_time[te],
                new_covariates = sim$C[te, ])
mean(sim$V[te] <= pred$fitted)

# Persistent component only
pred_B <- predict(fit, newY = sim$Y[te, ], new_subject = sim$subject[te],
                  new_visit_time = sim$visit_time[te],
                  new_covariates = sim$C[te, ], trajectory_only = TRUE)
mean(sim$V[te] <= pred_B$fitted)
```

```
## End(Not run)
```

`predict_population_lfqpca` *Population quantile curves at given visit times*

Description

Evaluates the population part $H(u)^\top \eta(s)$ of a fitted LFQPCA model at given visit times.

Usage

```
predict_population_lfqpca(object, new_visit_time)
```

Arguments

`object`	an object of class `lfqpca_fit`.
`new_visit_time`	visit times on the scale of the original `visit_time`.

Value

A matrix with one row per visit time and one column per grid point.

Examples

```
## Not run:
dat <- lfqpca_simulate("D1", I = 60, J = 5, replication = 1)
fit <- lfqpca(dat$Y, dat$subject, dat$visit_time, dat$grid, tau = 0.5,
              K_subject = 2, K_visit = 2, spline_df = 10,
              mean_time_degree = 2, max_iter = 60, tol = 1e-5, n_starts = 5,
              rq_method_functions = "fnb", rq_method_scores = "br",
              verbose = FALSE)

# Population medians at visit times -1, 0 and 1
pop <- predict_population_lfqpca(fit, new_visit_time = c(-1, 0, 1))
matplot(dat$grid, t(pop), type = "l", lty = 1:3, col = 1,
        xlab = "s", ylab = "population median")

## End(Not run)
```

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